

CodeAlpha Internship Project Report

Car Price Prediction Using Machine Learning

Abstract

This project develops a machine learning model to predict car selling prices using historical vehicle data. A Random Forest Regressor was trained and evaluated to provide accurate price predictions.

Objective

To build a regression model capable of estimating car prices based on vehicle characteristics such as year, fuel type, transmission, ownership history, and current market price.

Dataset Description

The dataset contains 301 records and 9 attributes: Car_Name, Year, Selling_Price, Present_Price, Driven_kms, Fuel_Type, Selling_type, Transmission, and Owner.

Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-Learn were used for data analysis, visualization, preprocessing, model building, and evaluation.

Methodology

The workflow included data loading, preprocessing, label encoding of categorical variables, train-test split, model training using Random Forest Regressor, prediction, and evaluation.

Model Evaluation

Results obtained: MAE = 0.5893 RMSE = 0.8847 R^2 Score = 0.9660 The model demonstrated excellent predictive capability.

Visualizations

The project includes Selling Price Distribution, Correlation Heatmap, Actual vs Predicted Prices, and Feature Importance Analysis.

Output Screenshots Section

Include screenshots of dataset preview, dataset information, model performance metrics, selling price distribution, heatmap, actual vs predicted graph, and feature importance graph.

Conclusion

The Random Forest model achieved approximately 96.6% prediction accuracy (R^2 score), indicating strong performance for car price prediction tasks.